Examples

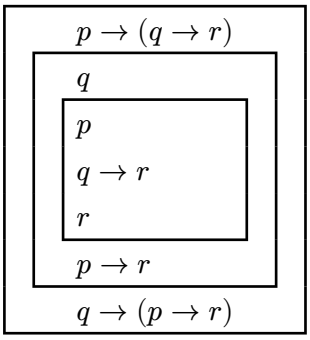
Default usage of `ded-nat-boxed`:

`nested-boxes`: `false`

$\vdash i \to, 2\text{-}6$

1.	$p \to (q \to r)$		**P**
2.		q	**P**
3.		$\quad p$	**P**
4.		$\quad q \to r$	$e \to, 3, 1$
5.		$\quad r$	$e \to, 2, 4$
6.		$p \to r$	$i \to, 3\text{-}5$
7.	$q \to (p \to r)$		$i \to, 2\text{-}6$

`nested-boxes`: `true`



1.	$p \to (q \to r)$	**P**
2.	q	**P**
3.	p	**P**
4.	$q \to r$	$e \to, 3, 1$
5.	r	$e \to, 2, 4$
6.	$p \to r$	$i \to, 3\text{-}5$
7.	$q \to (p \to r)$	$i \to, 2\text{-}6$

first line,
second line,
third line,
fourth line
$\vdash$ **fourth line**

1	1.	first line	**PR**
1	2.	second line	**PR**
1	3.	third line	**PR**
1	4.	**fourth line**	**PR**

Usage of `ded-nat-boxed` with overriden `style-dep`, `style-formula` and `style-rule` parameters:

first line,
second line,
third line,
fourth line
$\vdash$ **fourth line**

1	1.	first line	*P R*
1	2.	second line	*P R*
1	3.	third line	*P R*
1	4.	**fourth line**	*P R*

Deducción natural

Elegir 10 ejercicios del libro de Falguera y Vidal, de las páginas 318, 319, 320 y 321; y hacer su prueba con deducción natural.

1. pg. 318, ejercicios VI, I. 1)

   This is written using `ded-nat-boxed`. The coloring of the text is done by wrapping that function with `text(content, weight: "bold", fill: gradient.linear(red, blue))`, and the color of the box stroke is done with `stcolor: gradient.linear(red, blue)`.

$\forall x(Px) \land \forall x(Qx),$
$\forall x(Px \to Rx)$
$\vdash \forall x(Rx)$

1	1.	$\forall x(Px) \land \forall x(Qx)$	**PR**
2	2.	$\forall x(Px \to Rx)$	**PR**
1	3.	$\forall x(Px)$	**S 1**
1	4.	Pa	**IU 3**
2	5.	$Pa \to Ra$	**IU 2**
1,2	6.	Ra	**MP 4, 5**
1,2	7.	$\forall x(Rx)$	**GU 6**

   This is using `ded-nat`, and it is a repetition of the last one but without the boxing and without dependencies (inputting an array of 3 items).

1.	$\forall x(Px) \land \forall x(Qx)$	**PR**
2.	$\forall x(Px \to Rx)$	**PR**
3.	$\forall x(Px)$	**S 1**
4.	Pa	**IU 3**
5.	$Pa \to Ra$	**IU 2**
6.	Ra	**MP 4, 5**

7. $\forall x(Rx)$ **GU 6**

This is using `ded-nat-boxed`, without dependencies (inputting an array of 3 items) and without the premises and conclusion of the deduction automatically put over the lines.

1.	$\forall x(Px) \land \forall x(Qx)$	**PR**
2.	$\forall x(Px \to Rx)$	**PR**
3.	$\forall x(Px)$	**S 1**
4.	Pa	**IU 3**
5.	$Pa \to Ra$	**IU 2**
6.	Ra	**MP 4, 5**
7.	$\forall x(Rx)$	**GU 6**

2. pg. 321, ejercicios VI, I. 62)

1	1.	$\forall x(Sxb) \land \neg\forall y(Py \to Qby)$	**PR**
2	2.	$\forall x\forall y(Qxy \to \neg Qyx)$	**PR**
3	3.	$\neg\forall x(\neg Px) \to \forall y(Syb \to Qby)$	**Sup. RAA**
1	4.	$\neg\forall y(Py \to Qby)$	**S 1**
1	5.	$\exists y\neg(Py \to Qby)$	**EMC 4**
6	6.	$\neg(Pa \to Qba)$	**Sup. IE 5**
7	7.	$\neg(Pa \land \neg Qba)$	**Sup. RAA**
7	8.	$\neg Pa \lor \neg\neg Qba$	**DM 7**
9	9.	$\neg Pa$	**Sup. PC**
9	10.	$\neg Pa \lor Qba$	**Disy. 9**
	11.	$\neg Pa \to (\neg Pa \lor Qba)$	**PC 9-10**
12	12.	$\neg\neg Qba$	**Sup. PC**
12	13.	Qba	**DN 12**
12	14.	$\neg Pa \lor Qba$	**Disy. 13**
	15.	$\neg\neg Qba \to (\neg Pa \lor Qba)$	**PC 12-14**
7	16.	$\neg Pa \lor Qba$	**Dil. 8,11,15**
7	17.	$Pa \to Qba$	**IM 16**
6,7	18.	$(Pa \to Qba) \land \neg(Pa \to Qba)$	**Conj. 6, 17**
6	19.	$Pa \land \neg Qba$	**RAA 7-18**
6	20.	Pa	**S 19**
6	21.	$\exists x(Px)$	**GE 20**
6	22.	$\neg\forall x(\neg Px)$	**EMC 21**
3,6	23.	$\forall y(Syb \to Qby)$	**MP 3, 22**
3,6	24.	$Sab \to Qba$	**IU 23**
1	25.	$\forall x(Sxb)$	**S 1**

1	26.	Sab	**IU 25**
1,3,6	27.	Qba	**MP 24, 25**
6	28.	$\neg Qba$	**S 19**
1,3,6	29.	$Qba \vee \neg\exists y\neg(Py \to Qby)$	**Disy. 27**
1,3,6	30.	$\neg\exists y\neg(Py \to Qby)$	**MTP 28, 29**
1,3	31.	$\neg\exists y\neg(Py \to Qby)$	**IE 5, 6, 30**
1,3	32.	$\neg\exists y\neg(Py \to Qby) \wedge \exists y\neg(Py \to Qby)$	**Conj. 5, 31**
1	33.	$\neg(\neg\forall x(\neg Px) \to \forall y(Syb \to Qby))$	**RAA 3-32**

3. pg. 320, ejercicios VI, I. 51)

$$\neg(Mab \to (Sa \wedge Sb)),$$
$$\neg(\neg\exists xQx \vee \forall x\neg Rabx)$$
$$\vdash \forall x((Sa \wedge Sb) \to (Rabx \to \neg Qx))$$

1	**1.**	$\neg(Mab \to (Sa \wedge Sb))$	**PR**
2	**2.**	$\neg(\neg\exists xQx \vee \forall x\neg Rabx)$	**PR**
3	**3.**	$\neg(Mab \wedge \neg(Sa \wedge Sb))$	**Sup. RAA**
3	**4.**	$\neg Mab \vee \neg\neg(Sa \wedge Sb)$	**DM 3**
5	**5.**	$\neg Mab$	**Sup. PC**
5	**6.**	$\neg Mab \vee (Sa \wedge Sb)$	**Adj. 5**
5	**7.**	$Mab \to (Sa \wedge Sb)$	**IM 6**
	8.	$\neg Mab \to (Mab \to (Sa \wedge Sb))$	**PC 5-7**
9	**9.**	$\neg\neg(Sa \wedge Sb)$	**Sup. PC**
9	**10.**	$Sa \wedge Sb$	**DN 9**
9	**11.**	$\neg Mab \vee (Sa \wedge Sb)$	**Adj. 10**
9	**12.**	$Mab \to (Sa \wedge Sb)$	**IM 11**
	13.	$\neg\neg(Sa \wedge Sb) \to (Mab \to (Sa \wedge Sb))$	**PC 9-12**
3	**14.**	$Mab \to (Sa \wedge Sb)$	**Dil. 4, 8, 13**
1,3	**15.**	$\neg(Mab \to (Sa \wedge Sb)) \wedge (Mab \to (Sa \wedge Sb))$	**Conj. 1, 14**
1	**16.**	$Mab \wedge \neg(Sa \wedge Sb)$	**RAA 3-15**
17	**17.**	$S \wedge Sb$	**Sup. PC**
17	**18.**	$(S \wedge Sb) \vee (Rabc \to \neg Qc)$	**Adj. 17**
1	**19.**	$\neg(S \wedge Sb)$	**S 16**
1,17	**20.**	$Rabc \to \neg Qc$	**MTP 18, 19**
1	**21.**	$(Sa \wedge Sb) \to (Rabc \to \neg Qc)$	**PC 17-20**
1	**22.**	$\forall x((Sa \wedge Sb) \to (Rabx \to \neg Qx))$	**GU 21**

4. pg. 320, ejercicios VI, I. 41)

$$\forall x \forall y((Qyx \land Ryx) \rightarrow \neg Qxy),$$
$$Qab \land Pb$$
$$\vdash Rba \rightarrow \exists x(\neg Qbx \land Pb)$$

1	1.	$\forall x \forall y((Qyx \land Ryx) \rightarrow \neg Qxy)$	**PR**
2	2.	$Qab \land Pb$	**PR**
3	3.	Rba	**Sup. PC**
1	4.	$\forall y((Qya \land Rya) \rightarrow \neg Qay)$	**IU 1**
1	5.	$(Qba \land Rba) \rightarrow \neg Qab$	**IU 4**
2	6.	Qab	**S 2**
2	7.	$\neg\neg Qab$	**DN 6**
1,2	8.	$\neg(Qba \land Rba)$	**MT 5, 7**
1,2	9.	$\neg Qba \lor \neg Rba$	**DM 8**
3	10.	$\neg\neg Rba$	**DN 3**
1,2,3	11.	$\neg Qba$	**MTP 9, 10**
2	12.	Pb	**S 2**
1,2,3	13.	$\neg Qba \land Pb$	**Conj. 11, 12**
1,2,3	14.	$\exists x(\neg Qbx \land Pb)$	**GE 13**
1,2	15.	$Rba \rightarrow \exists x(\neg Qbx \land Pb)$	**PC 3-14**

5. pg. 319, ejercicios VI, I. 30)

$$\neg\exists x\exists y(\neg Txy \land \neg Tyx),$$
$$\forall x(Txa \to (Qa \land Ra)),$$
$$\neg\forall x(Tax)$$
$$\vdash \exists x(Qx \land Rx)$$

1	**1.**	$\neg\exists x\exists y(\neg Txy \land \neg Tyx)$	**PR**
2	**2.**	$\forall x(Txa \to (Qa \land Ra))$	**PR**
3	**3.**	$\neg\forall x(Tax)$	**PR**
3	**4.**	$\exists x\neg(Tax)$	**EMC 3**
5	**5.**	$\neg Tab$	**Sup. IE 4**
1	**6.**	$\forall x\neg\exists y(\neg Txy \land \neg Tyx)$	**EMC 1**
1	**7.**	$\neg\exists y(\neg Tby \land \neg Tyb)$	**IU 6**
1	**8.**	$\forall y\neg(\neg Tby \land \neg Tyb)$	**EMC 7**
1	**9.**	$\neg(\neg Tba \land \neg Tab)$	**IU 8**
1	**10.**	$\neg\neg Tba \lor \neg\neg Tab)$	**DM 9**
5	**11.**	$\neg\neg\neg Tab$	**DN 5**
1,5	**12.**	$\neg\neg Tba$	**MTP 10, 11**
1,5	**13.**	Tba	**DN 12**
2	**14.**	$Tba \to (Qa \land Ra)$	**IU 2**
1,2,5	**15.**	$Qa \land Ra$	**MP 13, 14**
1,2,5	**16.**	$\exists x(Qx \land Rx)$	**GE 15**
1,2,3	**17.**	$\exists x(Qx \land Rx)$	**IE 4, 5, 16**

6. pg. 319, ejercicios VI, I. 20)

$$\neg\exists x(Sx) \vee (Qa \wedge Ta),$$
$$\exists x(Qx \wedge Tx) \rightarrow \forall x(Rx),$$
$$Sa$$
$$\vdash Rb$$

1	**1.**	$\neg\exists x(Sx) \vee (Qa \wedge Ta)$	**PR**
2	**2.**	$\exists x(Qx \wedge Tx) \rightarrow \forall x(Rx)$	**PR**
3	**3.**	Sa	**PR**
3	**4.**	$\exists x(Sx)$	**GE 3**
3	**5.**	$\neg\neg\exists x(Sx)$	**DN 4**
1,3	**6.**	$Qa \wedge Ta$	**MTP 1, 5**
1,3	**7.**	$\exists x(Qx \wedge Tx)$	**GE 6**
1,2,3	**8.**	$\forall x(Rx)$	**MP 2, 7**
1,2,3	**9.**	Rb	**IU 8**

7. pg. 318, ejercicios VI, I. 10)

$$\forall x(Tx \rightarrow Qx),$$
$$\forall x\neg(Px \vee \neg Tx)$$
$$\vdash \exists x(\neg Px \wedge Qx)$$

1	**1.**	$\forall x(Tx \rightarrow Qx)$	**PR**
2	**2.**	$\forall x\neg(Px \vee \neg Tx)$	**PR**
2	**3.**	$\neg(Pa \vee \neg Ta)$	**IU 2**
2	**4.**	$\neg Pa \wedge \neg\neg Ta$	**DM 3**
2	**5.**	$\neg\neg Ta$	**S 4**
2	**6.**	Ta	**DN 5**
1	**7.**	$Ta \rightarrow Qa$	**IU 1**
1,2	**8.**	Qa	**MP 6, 7**
2	**9.**	$\neg Pa$	**S 4**
1,2	**10.**	$\neg Pa \wedge Qa$	**Conj. 8, 9**
1,2	**11.**	$\exists x(\neg Px \wedge Qx)$	**GE 10**

8. pg. 318, ejercicios VI, I. 5)

$$\forall x(Px) \to \forall x(Qx),\ \neg Qa \vdash \neg\forall x(Px)$$

1	**1.**	$\forall x(Px) \to \forall x(Qx)$	**PR**
2	**2.**	$\neg Qa$	**PR**
2	**3.**	$\exists x\neg(Qx)$	**GE 2**
2	**4.**	$\neg\forall x(Qx)$	**EMC 3**
1,2	**5.**	$\neg\forall x(Px)$	**MT 1, 4**

9. pg. 318, ejercicios VI, I. 6)

$$\forall x(Px \to Qx),\ \forall x(\neg Sx \to \neg Qx),\ \neg\forall x(Sx) \vdash \exists x(\neg Px)$$

1	**1.**	$\forall x(Px \to Qx)$	**PR**
2	**2.**	$\forall x(\neg Sx \to \neg Qx)$	**PR**
3	**3.**	$\neg\forall x(Sx)$	**PR**
3	**4.**	$\exists x\neg(Sx)$	**EMC 3**
5	**5.**	$\neg Sa$	**Sup. IE 4**
2	**6.**	$\neg Sa \to \neg Qa$	**IU 2**
2,5	**7.**	$\neg Qa$	**MP 5, 6**
1	**8.**	$Pa \to Qa$	**IU 1**
1,2,5	**9.**	$\neg Pa$	**MT 7, 8**
1,2,5	**10.**	$\exists x(\neg Px)$	**GE 9**
1,2,3	**11.**	$\exists x(\neg Px)$	**IE 4, 5, 10**

10. pg. 318, ejercicios VI, I. 7)

$$\forall x(Tx \to Mx),$$
$$\forall x\neg(Mx \wedge Rx),$$
$$\forall x(Tx \to (Px \to Rx))$$
$$\vdash \forall x(Tx \to \neg(Mx \to Px))$$

1	1.	$\forall x(Tx \to Mx)$	**PR**
2	2.	$\forall x\neg(Mx \wedge Rx)$	**PR**
3	3.	$\forall x(Tx \to (Px \to Rx))$	**PR**
4	4.	Ta	**Sup. PC**
1	5.	$Ta \to Ma$	**IU 1**
1,4	6.	Ma	**MP 4,5**
2	7.	$\neg(Ma \wedge Ra)$	**IU 2**
2	8.	$\neg Ma \vee \neg Ra$	**DM 7**
1,4	9.	$\neg\neg Ma$	**DN 6**
1,2,4	10.	$\neg Ra$	**MTP 8, 9**
3	11.	$Ta \to (Pa \to Ra)$	**IU 3**
3,4	12.	$Pa \to Ra$	**MP 4, 11**
1,2,3,4	13.	$\neg Pa$	**MT 10, 12**
14	14.	$Ma \to Pa$	**Sup. RAA**
1,4,14	15.	Pa	**MP 6, 14**
1,2,3,4,14	16.	$Pa \wedge \neg Pa$	**Conj. 13, 15**
1,2,3,4	17.	$\neg(Ma \to Pa)$	**RAA 14-16**
1,2,3	18.	$Ta \to \neg(Ma \to Pa)$	**PC 4-17**
1,2,3	19.	$\forall x(Tx \to \neg(Mx \to Px))$	**GU 18**